# TASK 02

# PHISHING EMAIL DETECTION AND AWARENESS REPORT

## 1. Introduction

Phishing is a type of cyber attack in which attackers send fake emails to steal personal information such as passwords, OTPs, and banking details. These emails may contain suspicious links or urgent messages to trick users.

In this project, I created a basic phishing email detector using Python. It checks the email subject and message for common phishing indicators.

## 2. Objective

The main objective of this project is to identify possible phishing indicators in email messages and create awareness about email security.

## 3. Tools Used

- Python
- Python IDLE
- Sample email messages
- Microsoft Word
- GitHub

## 4. Methodology

First, I created a Python program that accepts an email subject and message from the user.

The program checks for suspicious words such as "urgent", "immediately", "account suspended", and "password". It also checks for links and references to sensitive information.

If any of these indicators are found, the program displays a warning. Otherwise, it displays that no obvious phishing indicators were found.

## 5. Testing and Results

| Test Cases | Subject | Message | Output | Result |
|---|---|---|---|---|
| Normal Email | Meeting Schedule for tomorrow | Hello, our team meeting is scheduled tomorrow at 10 AM. Please review the project notes before joining. | No obvious phishing indicators found | The program did not identify any of its configured phishing indicators in this sample. |
| Suspicious Email | Urgent! Your account will be Suspended | Suspicious activity found. Immediately deactivate it. | Potential phishing indicators found – Urgent or threatening language detected. | The program identified urgent or threatening language in this sample. |

## 6. Common Phishing Indicators

- Urgent messages that pressure users to act immediately.
- Suspicious links and fake websites.
- Unexpected requests for passwords or OTPs.
- Fake sender addresses.
- Unexpected attachments.

## 7. Prevention Guidelines

- Do not click suspicious links.
- Check the sender's email address carefully.
- Never share passwords or OTPs through email.
- Verify unusual requests through official communication channels.
- Avoid opening unexpected attachments.
- Report suspicious emails to the appropriate security team.

## 8. Limitations

This project is a basic keyword-based detector. It cannot confirm whether an email is genuine or malicious. It may miss phishing emails that do not contain the configured keywords. Further improvements are required to make the system more reliable.

## 9. Conclusion

Through this project, I learned how to use Python to check email messages for possible phishing indicators. I tested the program using normal and suspicious email samples and observed the results. This project helped me understand basic phishing detection and email security awareness.